

Problem 1) Deriving the Initialization

$$\text{Var}[z^{(k)}] = \text{Var}\left[w^{(k)} h^{(k-1)}\right] = \text{Var}\left[\sum_{j=1}^{D_k} w_j^{(k)} h_j^{(k-1)}\right]$$

1) Explain assumptions that allow us to write:

$$\text{Var}\left[\sum_{j=1}^{D_k} w_j^{(k)} h_j^{(k-1)}\right] = D_k \text{Var}[w_k h_{k-1}]$$

$$\text{We have } z^{(k)} = \sum_{j=1}^{D_k} \underbrace{w_j^{(k)} h_j^{(k-1)}}_x \longrightarrow z^{(k)} = \sum_{j=1}^{D_k} x_j$$

The variance of sum formula: $\text{Var}\left(\sum_j x_j\right) = \sum_j \text{Var}(x_j) + 2 \sum_{i < j} \text{Cov}(x_i, x_j)$

here an assumption can be that the terms are uncorrelated so we assume their covariance is 0. $\text{Cov}(x_i, x_j) = 0$. That's why in the

$$\text{General formula: } \text{Var}\left(\sum_{i=1}^n x_i\right) = \sum_{i=1}^n \text{Var}(x_i) + 2 \sum_{i < j} \text{Cov}(x_i, x_j)$$

$\text{Var}\left[\sum_{j=1}^{D_k} w_j^{(k)} h_j^{(k-1)}\right]$, the covariance terms are ignored.

Also, as we sample each weight from $\mathcal{N}\left(0, \frac{2}{D_k}\right)$, weights are initialized from same distribution. weights are based on a normal distribution with mean 0 and variance σ^2 that is scaled by D_k . So we can

assume that all terms like $w_1 h_1, w_2 h_2, \dots$ have the same

variance and we can just write D_k once instead of writing it

$$\text{separate times} \rightarrow \text{Var}\left[\sum_{j=1}^{D_k} w_j^{(k)} h_j^{(k-1)}\right] = \underbrace{D_k}_{\substack{\# \text{ of units the layer feeds into} \\ \text{(inputs that go in neuron)}}} \text{Var}[w_k h_{k-1}]$$

2) We need to prove: $\text{Var}[w_k h_{k-1}] = \text{Var}[w_k] E[(h_{k-1})^2]$

$$\begin{cases} \text{Var}[AB] = \text{Var}[A] \text{Var}[B] + \text{Var}[A] E[B^2] + \text{Var}[B] E[A^2] \\ \text{Var}[B] = E[B^2] - E[B]^2 \end{cases}$$

- assumption about distribution of w

$$W \sim \text{Norm}[0, \sigma^2]$$

We assume when we initialize weights, they are from a normal distrib.

with mean of 0 $\rightarrow \underline{E[W] = 0}$

$$\text{Var}[AB] = \text{Var}[A] \text{Var}[B] + \text{Var}[A] E[B^2] + \text{Var}[B] E[A^2]$$

$$\underbrace{A = w_k \quad B = h_{k-1}}_{\rightarrow} \text{Var}[w_k h_{k-1}] = \frac{\overbrace{\text{Var}[w_k] \text{Var}[h_{k-1}] + \text{Var}[w_k] E[h_{k-1}^2]}^{\text{Var}[h_{k-1}] \cdot \underbrace{E[w_k]^2}_{=0}}}{\text{Var}[h_{k-1}] + E[h_{k-1}^2]}$$

$$\text{Var}[w_k h_{k-1}] = \text{Var}[w_k] \left(\underbrace{\text{Var}[h_{k-1}] + E[h_{k-1}^2]}_{\downarrow}$$

Given

$$\underline{\text{Var}[B] = E[B^2] - E[B]^2}$$

$$\text{Var}[h_{k-1}] = E[(h_{k-1})^2] - E[h_{k-1}]^2$$

$$\text{Var}[w_k h_{k-1}] = \text{Var}[w_k] (E[(h_{k-1})^2] - \underbrace{E[h_{k-1}]^2 + E[h_{k-1}]^2}_{\text{cancel out}})$$

$$\underline{\text{Var}[w_k h_{k-1}] = \text{Var}[w_k] E[h_{k-1}^2]}$$